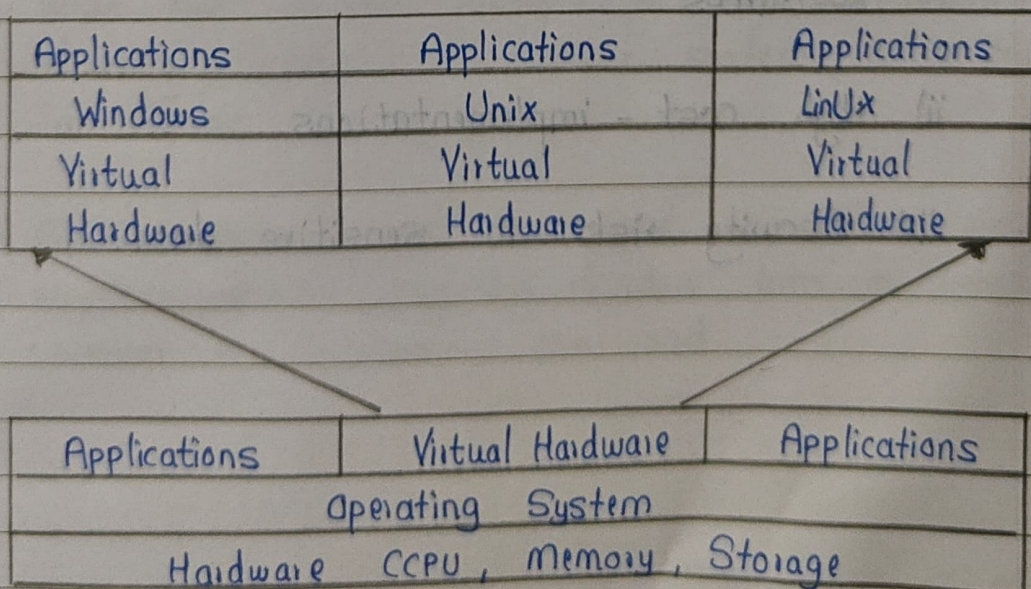


UNIT-III

Virtualization in Cloud Computing★ Virtualization -

- • Virtualization is broad term that refers to abstraction of resources across many aspects of computing.
- Virtualization is "creating of virtual (rather than actual) version of something, such as server, desktop, storage device, OS, etc."
 - In other words, virtualization is technique, which allows us to share single instance of resource or application among multiple customers & organization.
 - Diagram :-



Virtual machine.

- It does this by assigning a local name to physical storage and providing pointer to that physical resource when demanded.

■ Advantages of Virtualization:

- i) More flexible and efficient allocation of resources
- ii) It lowers the cost of IT infrastructure
- iii) Remote access and rapid scalability
- iv) High availability & disaster recovery
- v) Enables running multiple OS

■ Disadvantages of Virtualization:

- i) There is lack of availability of skilled resources
- ii) High cost - implementations
- iii) Security risk to sensitive data.

* Types of Virtualization -

→ I) Platform Virtualization -

- The creation of virtual machine using combination of hardware and software is known as platform virtualization.
- Platform virtualization is performed on given hardware platform by "host" software, which creates simulated computer environment for its "guest" software.

II) Server Virtualization -

- Server Virtualization is a software architecture that allows more than one server OS to run as guest on given physical server host.
- When the virtual machine software or Vmm is directly installed on server system, it is known as Server Virtualization.
- Server virtualization is done because a single physical server can be divided into multiple server for balancing the load.

III) Operating System Virtualization -

- It is a server-virtualization method where the kernel of OS allows for multiple isolated user-space instances, instead of one.
- When the virtual machine software or Vmm is installed on Host OS instead of directly on hardware system is known as OS virtualization.
- It is mainly used for testing applications on different platform of OS.

IV) Para-Virtualization -

- It is a type of virtualization in which guest OS is recompiled, installed inside vm, operated on top of hypervisor program running on host OS.
- It refers to the communication between guest OS and hypervisor to improve performance and efficiency.

* Comparison -

Full Virtualization

- i) It relies upon binary translation to trap & to virtualize certain sensitive instructions

Eg, VMware

- ii) It doesn't need to modify host OS

- iii) Normal instructions can run directly on host OS

- iv) It uses binary translation & direct execution

- v) Performance is good

- vi) Guest software does not need any modification since underlying hardware is fully simulated

Para Virtualization

- i) It refers to communication between guest OS and hypervisor to improve performance & efficiency

Eg, Xen architecture

- ii) It involves modification of OS Kernel.

- iii) Para-virtualized systems won't run on native hardware.

- iv) It uses hyper-calls

- v) Performance is better in certain cases

- vi) Hardware is not simulated & guest software run their own isolated domains.

★ Storage Virtualization -

- • Storage Virtualization is the process of grouping physical storage from multiple network storage devices so that it looks like single storage device
- Storage virtualization is also implemented by software applications
 - Storage virtualization is mainly done for back-up and recovery purposes.

■ Benefits :

- i) Data is stored in more convenient locations away from host
- ii) Lower total cost of ownership
- iii) Improved physical resource utilization.

★ Network Virtualization -

- • Network virtualization facilitates running of multiple networks, multiple consumers over shared substrate
- It is a method which combines available resources by splitting up bandwidth into channels & is assigned to device or user in real time.

- By adopting network virtualization, managing the network will be easier job & less time-consuming for network administrators.
- Components of virtual network consist of:
 - i) VLANs and VMS
 - ii) Network storage devices
 - iii) Network elements eg, Firewall
 - iv) Network mobile elements eg, tablets, mobiles
 - v) Network media eg, Fibre channels.

* CPU Virtualization -

- • Certain processors such as Intel VT provide hardware assistance for CPU Virtualization.
- By adopting CPU virtualization, two separate CPUs resemble a single CPU, i.e. two systems running in single system.
- The main objective of CPU virtualization is to make a CPU function similar to two separate CPUs.
- CPU virtualization allows the users to run different OS simultaneously
- Eg, Apple Mac can be virtualized to run Windows as well.

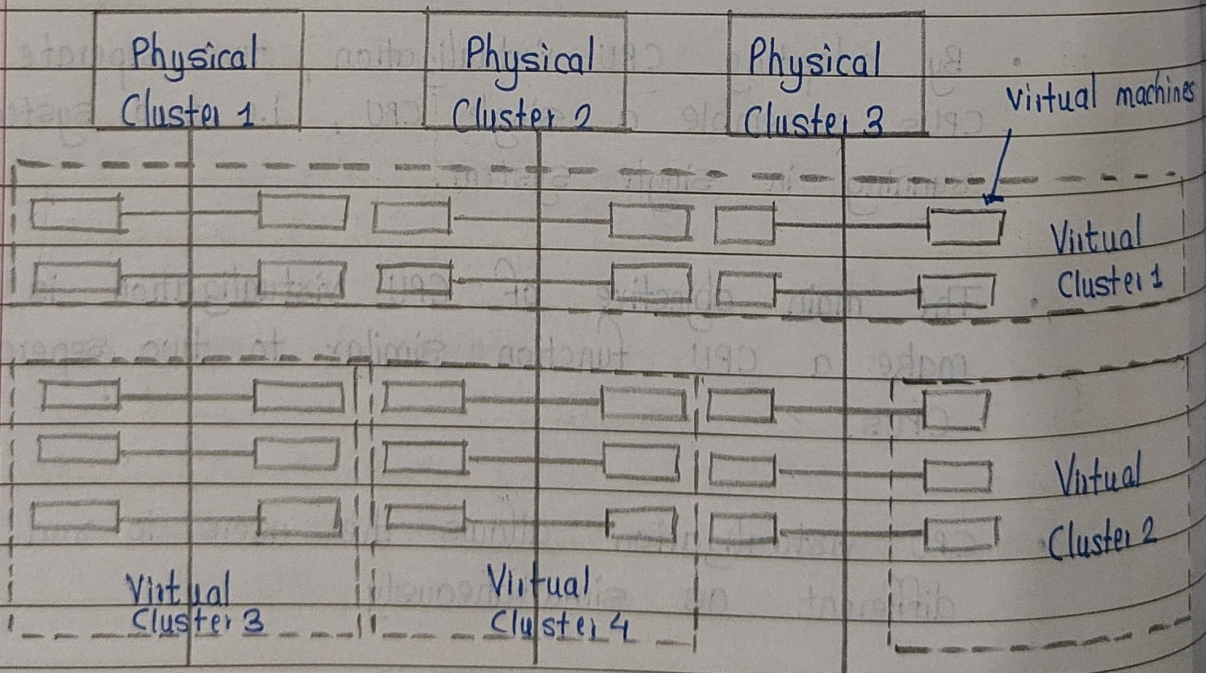
★ Virtual Clustering -

→ • Virtual cluster refers to set of interconnected VMs that are configured to work together as cohesive computing environment.

• Virtual clusters are built with VMs installed at distributed servers from one or more physical clusters

• VMs in virtual cluster are interconnected logically by virtual network across several physical networks.

• Diagram -



- Each virtual cluster is formed with physical machines or VM hosted by multiple physical clusters.

■ Characteristics of Virtual Cluster:

- i) VM runs with guest OS.
Host OS and VM OS are different but it manages resources in physical machine.
- ii) VM can be replicated in multiple servers & it supports distributed parallelism, fault tolerance.
- iii) No. of nodes of virtual cluster may change accordingly.
- iv) If VM fails, it cannot affect host machine.

■ Virtual cluster can be managed by four ways:

- i) We can use guest-based manager, eg. Linux.
- ii) We can use host-based manager, eg. VMware HA.
- iii) We can use independent cluster manager.
- iv) We can use integrated cluster manager.

* Virtualization in Grid -

→ • Grid computing is a distributed computing system where a group of computers are connected to create & work as one large virtual ~~real~~ computing power, storage, service, etc.

- Primary focus in Grid computing lies in secure resource sharing in terms of access to computers, software in dynamic environment.
- Grid has many systems in a network & hence multiple people can have ownership.
- Virtualization helps in providing cloud, better security.
- Grid computing is more economical as it splits work and distributes it over network on computers, increasing efficiency.
- Eg, Running multiple window servers (vm) on an Intel box.

* Comparison -

Grid Computing

i) It follows a distributed computing architecture

ii) Scalability is ~~high~~ normal

iii) Grid computing is less flexible than CC

iv) Grid operates as a decentralized management system

v) Grid computing is application oriented

vi) GC uses systems like distributed computing, distributed information, etc

vii) Resources are managed by different organizations & individuals

viii) Eg, BOINC, SETI@home

Cloud computing

i) It follows client-server computing architecture

ii) Scalability is ~~low~~ high

iii) Cloud computing is more flexible than GC

iv) Cloud operates as a centralized management system

v) Cloud computing is service oriented.

vi) CC uses services like IaaS, SaaS, PaaS.

vii) Resources are managed by single cloud provider

viii) Eg, AWS, Microsoft Azure.

★ Hypervisor & Types -

→ • Virtualization is commonly hypervisor based.

- The hypervisor isolates OS and applications from underlying computer hardware.
- Due to this, host machine can run multiple VM as guests, that share system's physical resources, such as memory space, bandwidth, etc.

▪ Type-1 hypervisor:

- They are also called as bare-metal hypervisors.
- They run directly on top of host system hardware.
- Type-1 hypervisor offer high availability & resource management.
- Their direct access to system hardware enables better performance, scalability, etc.
- Eg, Microsoft Hyper-V, VMware ESX;

■ Type - 2 hypervisor :

- They are also known as Hosted hypervisor.
- They are installed on top of host OS rather than sitting directly on top of hardware as type - 1 does
- Convenience of known host OS can ease system configuration and management tasks.
- However, addition of host OS layer can limit performance & expose possible OS security flaws.
- Eg, VMware Workstation, Virtual PC, etc.